

Sanjay Noolu

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SUMMARY

Senior Full Stack Engineer with 7+ years of expertise architecting high-throughput distributed systems, cloud-native microservices, and modern AI infrastructure. Deeply proficient across the full stack, including Java/Spring Boot services, React/Angular front-ends, and scalable API design, with extensive practical experience implementing production-grade AI automation, RAG pipelines, agentic tool-calling systems, and intelligent orchestration with built-in security guardrails. Proven record of optimizing JVM performance, streamlining CI/CD workflows, and managing complex relational and NoSQL data layers for enterprise organizations.

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, Groovy

AI & LLM: LangChain, LangGraph, Prompt Engineering, Agentic Workflows, Multi-Agent Systems, NLP, Multimodal Models, Claude Code

RAG & Vector Search: RAG, Vector Databases (Pinecone, ChromaDB), Semantic Search, Vector Embeddings, MCP

Backend & Data: FastAPI, Spring Boot, RESTful APIs, Microservices, OAuth, SQL (PostgreSQL, MySQL, Oracle), NoSQL Databases (MongoDB)

Testing & Evaluation: Pytest, Hallucination Detection, RAG Metrics (Recall@K, MRR), Unit Testing

Web: React, Angular, HTML/CSS, Bootstrap, Node.js

Cloud & DevOps: AWS (EC2, S3, Lambda, Bedrock), Azure, Docker, Kubernetes, Git, Jenkins, CI/CD

EXPERIENCE

Senior Full Stack Engineer – Fidelity Investments, Dallas, TX

Nov 2022 – Present

- Currently spearheading architecture and development of an in-house enterprise application to replace the company's legacy Oracle Siebel CRM platform end-to-end, projected to save the organization \$500K in third-party licensing and support costs.
- Led peer code reviews and architectural discussions, establishing strict coding standards that reduced post-release production defects.
- Architected and scaled distributed microservices using Java and Spring Boot for an enterprise pension and retirement platform, managing complex computation logic to ensure calculation precision for high-volume financial payouts.
- Designed and optimized secure RESTful APIs via Spring Boot, implementing advanced data structures and query caching to significantly reduce endpoint latency.
- Orchestrated cloud-native container build and release configurations using Docker and Kubernetes (AKS) to ensure repeatable, seamless production deployments.
- Authored rigorous automated test suites utilizing Mockito and JUnit 5, elevating total test coverage to 80% and ensuring code extensibility.
- Identified and resolved system bottlenecks by analyzing JVM memory allocation patterns, garbage collection cycles, and thread contention metrics via Datadog.

Associate Software Developer – Citi Bank, Dallas, TX

Nov 2019 – Nov 2022

- Engineered core backend services for an enterprise digital banking platform using functional Java patterns, Stream APIs, and modern multi-threading constructs to securely and efficiently process high-volume web and mobile client transactions.
- Implemented an API gateway (Azure APIM) to control traffic between browser and mobile APIs, ensuring security and performance.
- Improved release quality by writing automated BDD/TDD validation scripts with Cucumber, integrated directly into Jenkins CI/CD pipelines.
- Architected dual-tier distributed data patterns utilizing Gemfire for in-memory caching and MongoDB as a low-latency NoSQL operational data layer, optimizing high-frequency read loops.
- Built and supported a customer-facing Angular application for the Credit Offers department, used directly by customers

to view personalized promotional offers and manage activation status.

- Implemented Log4j-based logging and used Splunk to monitor production APIs daily, proactively identifying and resolving issues in collaboration with the production support team.

Software Developer – AT&T, Atlanta, GA

April 2019 – Nov 2019

- Managed the complete engineering lifecycle of an enterprise Interactive Voice Response (IVR) application on the Avaya platform, leading call-flow system design and BPMN process modeling via Camunda.
- Built high-throughput Java microservices via Spring Boot to fetch and transform transactional business data from secure, on-premises relational databases (MySQL).
- Integrated Apache Kafka message brokers to enable loose coupling, event-driven processing, and resilient service-to-service messaging.

Software Developer – SAS Institute, Raleigh, NC

July 2018 – March 2019

- Migrated legacy marketing systems to a modern cloud architecture, developing scalable Spring Boot microservices deployed on AWS EC2 infrastructure.
- Refactored the persistence layer from legacy JDBC references into optimized Spring Data JPA entities and custom MySQL queries, including new entity classes for the migrated data model.

PROJECTS

Enterprise Q&A Assistant with Corrective RAG (ConfluenceAssist)

- Engineered production-grade QA system serving internal knowledge queries with grounded, hallucination-resistant responses using corrective RAG architecture.
- Implemented LangGraph state machine to evaluate retrieval quality, auto-rewrite failed searches, and validate answers against source citations.
- Built security guardrails detecting and mitigating prompt injection attempts, and classifying query intent (GREETING, INTERNAL_QUERY, OFF_TOPIC).
- Deployed a Python/FastAPI backend with a comprehensive evaluation suite: LLM-as-a-Judge, Recall@K, MRR metrics, and answer quality assessment (Faithfulness, Correctness).

Text-to-SQL Agent using Amazon Bedrock

- Developed a Python-based, LLM-powered conversational agent translating natural language queries into executable SQL for enterprise databases.
- Designed multi-step prompt engineering with schema-aware context injection and self-healing error loops for automatic SQL correction.
- Implemented retrieval-augmented generation (RAG) to inject database metadata, reducing hallucinations and improving query accuracy.
- Enabled non-technical users to access complex data insights through a scalable chatbot interface deployed on AWS.

Voice-First AI Travel Booking Agent

- Architected a voice-first conversational booking agent that orchestrates real-time flight, hotel, and car reservations end-to-end through natural spoken dialogue, integrating low-latency streaming STT/TTS with a dual-provider LLM.
- Built a Model Context Protocol (MCP) client against Sabre's GDS Sandbox to power live flight shopping, hotel search/rate retrieval, and reservation management, silently managing stateful `conversationId` sessions under a stateless voice conversation model to prevent LLM hallucination.
- Implemented parallel tool-calling in Python within the LLM execution loop (`ThreadPoolExecutor`) to run concurrent Sabre GDS, hotel, and weather lookups, cutting Turn-1 voice search latency by 50% and eliminating conversational timeout failures.
- Designed a MySQL-backed conversational memory layer feeding recent session context into the prompt tail, enabling the voice agent to resolve multi-turn references (e.g., follow-up questions about a prior search) without breaking dialogue flow.

EDUCATION

The University of Texas at Arlington (UTA), Arlington, TX

May 2018

Master of Science (M.S.), Computer Science

Jawaharlal Nehru Technological University (K), Visakhapatnam, India

April 2016

Bachelor of Technology (B.Tech.), Information Technology